

Models and mechanism

- Keystone goal: **Prediction** ("What would happen if...")

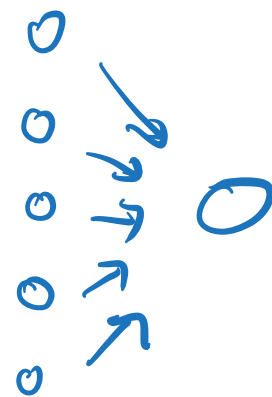
if **A** is increased, then...



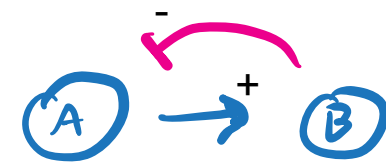
- Keystone challenge: Biological systems are **complex** (not just big)



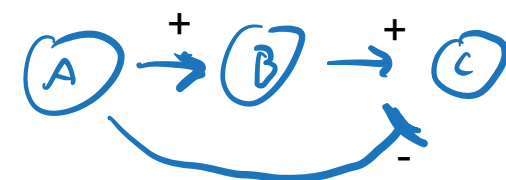
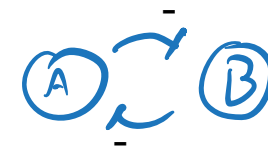
not complex



not complex

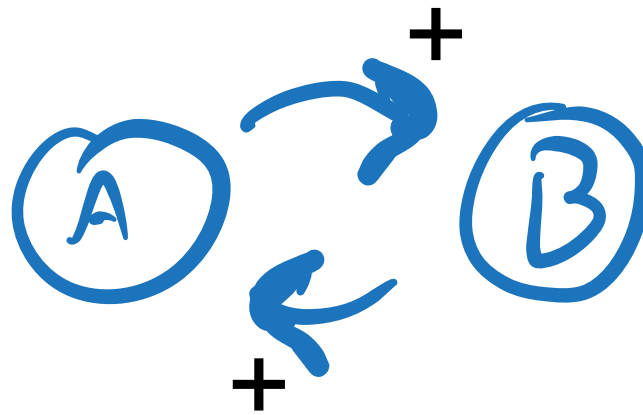


complex!



- Keystone tool: Dynamical systems, control theory, (the differential equation,) ...

stress

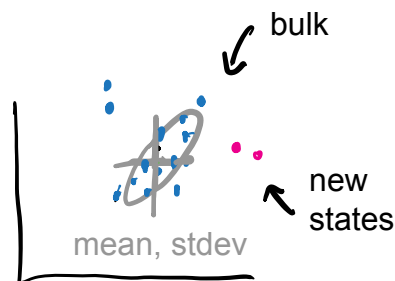


breathing
rate

the goals, challenges, systems, tools and approaches of today

Data and analytics

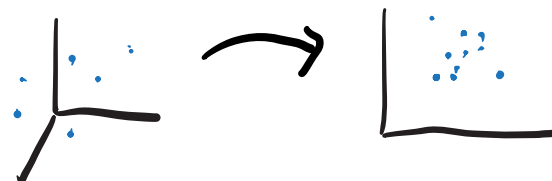
- Keystone goal: **Discover the hidden** (new states, cell types, disease classes,...)



- Keystone challenge: Biological data is not just big, it's **high-dimensional**.

30000 genes
1000 cells

10^4 microbial species
20 patients



- Keystone tool: Clustering, regression, dimension-reduction... coding

Models and mechanism

- Keystone goal: **Prediction** ("What would happen if...")

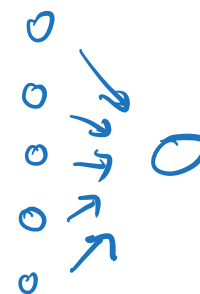
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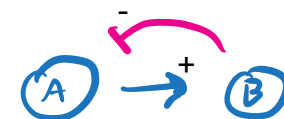
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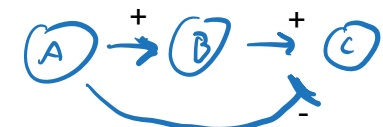
not complex



not complex



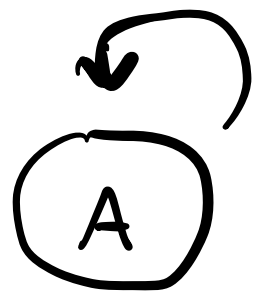
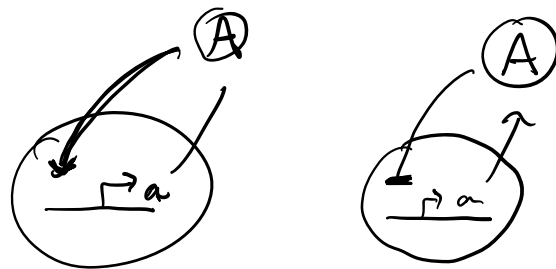
complex!



- Keystone tool: Dynamical systems, control theory, (the differential equation,) ...

Complex biological systems and “systems thinking”

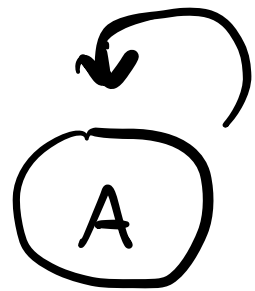
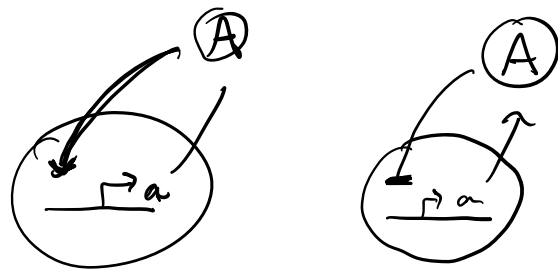
→ Activation
⊥ Repression



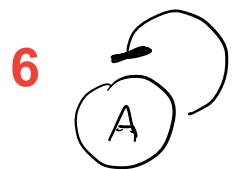
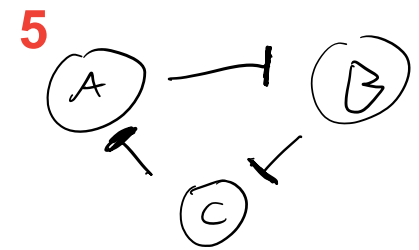
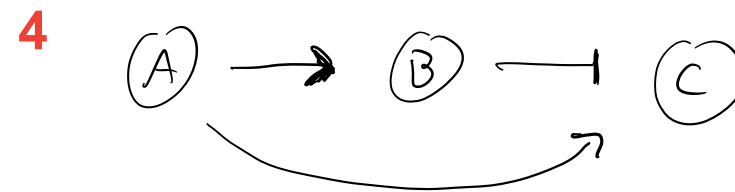
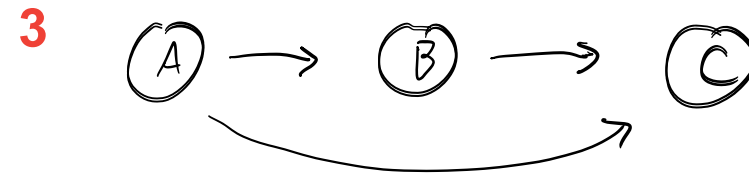
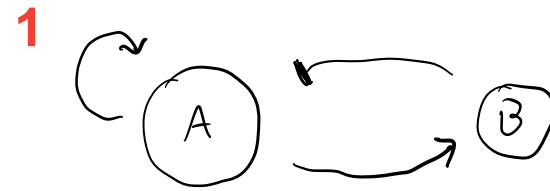
Positive feedback

- With positive feedback, the curve showing steady-state protein as a function of activator TF should move up.
- Does the curve move up by:
 - Shifting the EC50 to the left?
 - Moving up at high-C?
 - Moving up at low-C?
 - Combination of all 3?

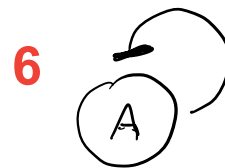
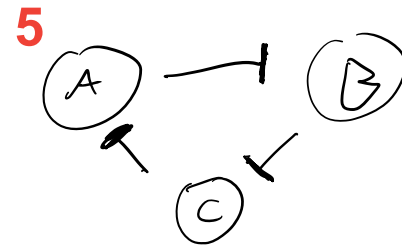
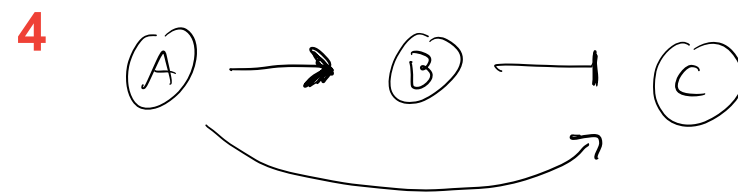
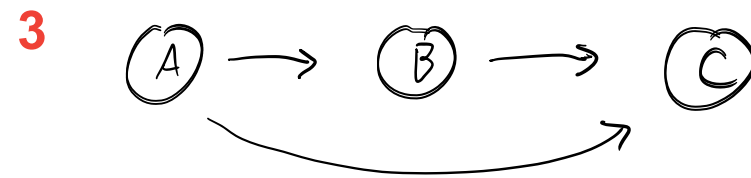
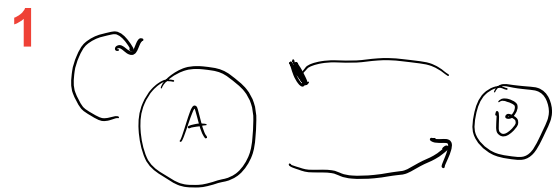
→ Activation
 T Repression



Positive feedback



In-class activity: Matching



- Direct negative feedback 6
- Indirect negative feedback 7
- Double-negative feedback 2
- Fast positive and slow negative feedback 1
- Incoherent feedforward 4
- Coherent feedforward 3
- The Repressilator 5